Math 241, F12 **Name:** Answer

Quiz 5 **SSN:** xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Let $\mathbf{r}(t) =< 5t,\ 2\sin t,\ 2\cos t >$. Find the curvature of the curve using the formula $\kappa = \dfrac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}$.

$$r'(t) = <5, 2\cos t, -2\sin t> \Rightarrow |r'(t)| = \sqrt{5^2 + (2\cos t)^2 + (-2\sin t)^2}$$
$$= \sqrt{25 + 4(\cos^2 t + \sin^2 t)} = \sqrt{29}$$

$$r''(t) = <0, -2\sin t, -2\cos t>$$

$$r'(t) \times r''(t) = \begin{vmatrix} i & j & k \\ 5 & 2\cos t & -2\sin t \\ 0 & -2\sin t & -2\cos t \end{vmatrix} = (-4\cos^2 t - 4\sin^2 t)i - (-10\cos t)j + (-10\sin t)k$$
$$= -4i + 10\cos t\, j - 10\sin t\, k.$$

$$\Rightarrow |r'(t) \times r''(t)| = \sqrt{(-4)^2 + (10\cos t)^2 + (-10\sin t)^2} = \sqrt{16 + 100} = \sqrt{116} = 2\sqrt{29}$$

Thus
$$k = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3} = \frac{2\sqrt{29}}{29^{3/2}} = \frac{2}{29}$$

Problem 2. (5 points) Find the velocity vector $\mathbf{v}(t)$ and the position vectors $\mathbf{r}(t)$ of a particle that has $\mathbf{a}(t) =< 1, 0, 2 >$, $\mathbf{v}(0) =< 0, 0, 1 >$ and $\mathbf{r}(0) =< 1, 0, 0 >$.

$$V(t) = \int a(t)\,dt$$
$$= \int <1, 0, 2>\,dt = <t, 0, 2t> + C_1$$

$$V(0) = <0, 0, 1> \Rightarrow <0, 0, 0> + C_1 = <0, 0, 1> \Rightarrow C_1 = <0, 0, 1>$$

$$\Rightarrow V(t) = <t, 0, 2t> + <0, 0, 1>$$
$$= <t, 0, 2t+1>$$

$$r(t) = \int V(t)\,dt = \int <t, 0, 2t+1>\,dt = <\frac{t^2}{2}, 0, t^2+t> + C_2$$

$$r(0) = <1, 0, 0> \Rightarrow <0, 0, 0> + C_2 = <1, 0, 0> \Rightarrow C_2 = <1, 0, 0>$$

Thus $$r(t) = <\frac{t^2}{2}, 0, t^2+t> + <1, 0, 0>$$
$$= <\frac{t^2}{2} + 1, 0, t^2+t>$$